

ASWAN INTL, Egypt

WMO: 624140

Lat: 23.964N

Lon: 32.820E

Elev: 202

StdP: 98.92

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.1	9.3	-9.2	1.8	22.4	-7.2	2.1	23.0	9.1	19.7	8.3	18.9	2.9	10	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	44.3	20.9	43.2	20.7	42.2	20.4	22.6	40.6	21.9	40.4	21.3	39.9	4.7	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	11.7	31.5	14.4	10.5	32.8	13.2	9.7	33.2	67.4	40.7	64.7	40.3	62.6	40.0	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	8.0	7.2	DB	5.3	46.3	1.9	0.9	3.9	46.9	2.8	47.4	1.8	47.9	0.4	48.6	(4)
(5)			WB	1.5	24.6	1.0	1.0	0.8	25.3	0.2	25.9	-0.3	26.5	-1.1	27.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.3	16.9	19.0	23.1	28.0	32.1	34.3	35.0	35.0	32.9	29.3	23.1	18.1
	DBStd	7.19	3.12	3.59	3.71	3.67	2.99	2.07	1.68	1.56	1.87	3.15	3.21	3.01
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	143	67	30	4	0	0	0	0	0	0	0	3	40
	CDD10.0	6304	215	253	406	538	685	728	774	774	688	598	393	253
	CDD18.3	3406	23	49	151	289	427	478	516	516	438	339	146	34
	CDH23.3	52156	243	583	1785	3989	6740	8031	8871	8743	6882	4538	1439	312
CDH26.7	34404	61	232	858	2401	4531	5679	6400	6270	4591	2694	602	86	
Wind	WSAvg	3.9	3.5	3.7	3.8	4.0	4.1	4.3	4.2	4.0	3.8	3.6	3.5	
Precipitation	PrecAvg	2	0	0	1	0	0	0	0	0	0	0	0	0
	PrecMax	18	3	0	15	9	3	0	0	0	3	9	1	2
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	4	0	0	3	2	1	0	0	0	1	1	0	0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.9	35.2	38.9	42.8	45.3	46.0	45.6	45.0	43.9	42.2	36.8	31.8
		MCWB	17.3	17.6	19.2	19.8	20.3	20.6	21.8	21.6	21.4	20.3	19.4	18.4
	2%	DB	28.2	32.1	36.1	40.8	43.2	44.2	44.0	43.6	42.1	40.2	34.2	29.1
		MCWB	16.5	16.3	17.6	19.4	20.3	20.5	21.0	21.4	20.8	20.4	18.8	17.0
	5%	DB	26.1	29.8	33.9	38.8	41.8	43.0	42.8	42.3	40.9	38.3	32.1	26.8
		MCWB	14.9	15.2	16.6	18.3	19.8	20.1	20.7	20.9	20.4	20.0	17.9	15.8
	10%	DB	23.9	26.9	31.4	36.8	40.1	41.5	41.8	41.2	39.8	36.8	30.1	24.6
		MCWB	13.6	13.9	15.4	17.3	19.1	19.6	20.3	20.5	20.0	19.6	17.0	14.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.2	19.4	20.6	21.6	22.8	22.4	23.7	23.4	22.7	22.7	21.5	20.1
		MCDB	28.2	31.8	35.1	39.7	42.0	41.7	41.5	41.6	40.8	37.6	33.0	28.8
	2%	WB	17.5	17.2	18.6	20.0	21.3	21.4	22.1	22.3	21.7	21.5	19.9	17.6
		MCDB	26.6	30.1	34.7	38.5	40.7	41.7	41.5	41.3	39.7	37.5	31.8	27.8
	5%	WB	15.3	15.4	17.1	18.7	20.3	20.7	21.4	21.6	21.1	20.6	18.6	16.0
		MCDB	25.2	28.7	32.1	37.1	39.9	41.0	40.9	40.7	39.1	36.9	30.9	25.9
	10%	WB	13.7	14.0	15.8	17.8	19.4	20.1	20.7	21.0	20.4	19.8	17.3	14.6
		MCDB	23.3	26.2	30.7	35.8	38.8	40.1	40.1	40.0	38.2	35.7	29.2	24.2
Mean Daily Temperature Range	5% DB	MDBR	12.9	13.7	14.4	14.8	14.3	14.3	13.5	13.4	13.8	13.5	13.1	12.6
		MCDBR	15.1	16.7	16.6	16.8	15.8	15.4	14.3	13.9	14.7	14.9	14.7	14.2
		MCWBR	8.4	8.2	8.0	7.3	6.8	6.3	5.7	5.3	5.9	6.2	7.0	8.1
	5% WB	MCDBR	14.7	16.3	15.9	16.2	15.1	14.6	13.7	13.5	14.1	14.0	14.0	14.2
		MCWBR	8.9	8.6	8.1	7.6	6.9	6.4	6.0	5.6	6.1	6.4	7.6	8.4
Clear-Sky Solar Irradiance	taub	0.370	0.407	0.449	0.534	0.552	0.486	0.506	0.525	0.494	0.473	0.393	0.371	
	taud	2.200	2.062	1.936	1.747	1.726	1.899	1.882	1.849	1.906	1.949	2.150	2.197	
	Ebn at Noon	880	870	851	786	768	814	798	783	799	793	846	861	
	Edh at Noon	129	159	189	233	237	198	201	208	192	175	135	125	
All-Sky Solar Radiation	RadAvg	4.83	5.72	6.80	7.54	7.90	8.15	7.91	7.48	6.87	5.90	5.03	4.54	
	RadStd	0.15	0.17	0.20	0.18	0.19	0.10	0.12	0.13	0.15	0.09	0.09	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (1 neighbor)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.70	+0.45	N/A	N/A	+0.30	N/A	N/A	-57	+256	+204

Nomenclature: See separate page